

Karthik Vinayan

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Summary

Building AI at Clueso (YC W23). Previously built the backend for a production AI cloud automation platform – multi-agent orchestrator, knowledge graph infra, MCP, semantic memory, all from zero.

Experience

Applied AI Engineer

May 2026 – Present

Clueso

- Building AI at Clueso (YC W23). AI that turns screen recordings into polished product videos and step-by-step docs.

Founding AI Engineer

Jun 2024 – Apr 2026

Omni RPA Inc / Agentic Solutions Pvt Ltd

- MCP server and client from scratch. Built while the protocol was 5 weeks old. Tool gating with prerequisite chains, dynamic schema generation, attribute matrix filtering, mandatory reason parameter on every pricing call to kill speculative tool use and create an audit trail.
- DAG-based multi-agent orchestrator. 8 agent types (pricing-researcher, memory-retriever, cost-analyzer, constraint-validator, recommendation-builder, and others). Topological sort, cycle detection, parallel tier execution, WebSocket streaming. Users see a live DAG panel with agents moving through pending/running/completed.
- Knowledge graph and RAG stack. 7-stage NATS pipeline for document ingestion with 85–95% of entity extraction handled locally via GLiNER ONNX (no LLM API calls). Neo4j to FalkorDB migration, hybrid retrieval on Weaviate/gRPC, two-layer semantic cache to avoid redundant LLM calls.
- Semantic memory on PostgreSQL with pgvector. Hybrid ranking by similarity + recency + relevance, rolling summarization, personalized welcome prompts for returning users. Agents remember past conversations.
- Onboarded first enterprise client pre-launch. Hands-on usage, bug filing, iterated on agent behavior and UX across 32 deployment workflows (multi-region DR, HIPAA/PCI-DSS, EKS, ML platforms).
- Own 3–5 repos end-to-end, primary on-call. 2hr worst-case recovery. Lead work across frontend, data, Java backend, and DevOps in a 10-person remote team.

Research Intern

Oct 2023 – Dec 2023

Digital University of Kerala

- Built real-time crop ripeness detection system using YOLOv8 fine-tuned on proprietary agricultural dataset; deployed to production.

Open Source & Projects

podspawn – One-command dev environments, locally or over SSH.

juno – Local coding agent in the terminal.

Tenso – Postman alternative built with Tauri 2.0 and SolidJS.

hn – Hacker News client.

Skills & Certifications

Languages: Python, Go, Rust, TypeScript, SQL, Java

Backend: FastAPI, WebSockets, NATS JetStream, Event-Driven Architecture

AI & ML: Multi-Agent Orchestration, MCP Protocol, RAG, Semantic Caching, LLM Tool Calling, Eval Frameworks, pgvector, vLLM, llama.cpp

Databases: FalkorDB, Neo4j, Weaviate, Supabase (Postgres), Redis

Cloud & DevOps: AWS (CCP), Azure (AZ-104), GCP, Docker, Kubernetes, OpenTelemetry

Publication

Automated Medicine Delivery System for Hospitals – Springer LNNS, 2025

Led team building Raspberry Pi autonomous robot with obstacle detection, path following, and RFID room ID. DOI: 10.1007/978-981-96-3652-5_40

Education

Vellore Institute of Technology – B.Tech CSE, AI & ML Specialization

Sep 2021 – Aug 2025